Mostly Noise: Privacy-Budget Allocation and Trust-Based Robustness in Differentially Private Federated Drone Detection

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Abstract*—How to spend a privacy budget over training rounds, and how to aggregate updates when some clients are malicious, are usually studied separately and at weak privacy. We study both on one differentially private (DP) federated infrared drone detector at worst-client $\varepsilon = 0.79$–2.72 ($\delta = 10^{-5}$, frame-level), with pre-registered rules, 20 paired seeds and Holm correction. At $\varepsilon = 1.86$ an honest update's cosine with a clean server update is 0.0075. A fixed schedule with less noise in later rounds beats uniform allocation by $+0.0139$ mAP@0.5 [$+0.0076$, $+0.0202$] ($+23.5\%$, 16/20 seeds). Reinforcement learning, Bayesian optimisation and a constrained learned policy add nothing. FLTrust loses 75% of accuracy without attack. A noise-aware rescaling restores it, but a rule-aware attacker drives it to 0.0005 mAP@0.5, and capping each client's trust weight at 1/8 still leaves 0.0058. Limitation: one primary dataset, frame-level privacy, low absolute accuracy (uniform 0.059 mAP@0.5). Central claim: *at strong privacy an honest DP update is almost entirely noise; consequently, all the allocation gain we could measure comes from placing less noise in later rounds, which a fixed schedule already captures, while trust-based robust aggregation either drowns the honest signal or hands control to a coherent attacker.

***Index Terms*—Differential privacy, federated learning, privacy-budget allocation, Byzantine robustness, FLTrust, adaptive attacks, object detection, pre-registration.**

I. Introduction

Consider eight operators of counter-drone infrared cameras: airports, border posts, and sites near buildings, hills and cloud-heavy sky. Each holds footage that cannot leave its premises, and each wants a better drone detector than its own data can train. Federated learning (FL) [1] keeps the footage local. DP-SGD [2] bounds what the shared model reveals about any single frame. Two questions then decide whether the system is worth deploying. First, the privacy budget is fixed, so how should it be spent over training rounds? Second, some operators may be compromised, so which aggregation rule protects the model?

Both questions have active literatures, but they are studied apart and mostly at weak privacy. Time-adaptive budget allocation [3] reports gains at $\varepsilon = 10$–30 from three-run averages without significance tests. Byzantine-robust aggregation rules such as Krum [4], the coordinate-wise median [5] and FLTrust [6] are designed and evaluated mostly without DP noise. Guerraoui et al. [7] show theoretically that the two kinds of guarantee interact badly as the model dimension grows. What practitioners lack is a controlled, statistically careful measurement of both choices on one realistic task at the privacy levels a regulator would accept.

This paper provides that measurement. Its central claim is: *at strong privacy an honest DP update is almost entirely noise; consequently, all the allocation gain we could measure comes from placing less noise in later rounds, which a fixed schedule already captures, while trust-based robust aggregation either drowns the honest signal or hands control to a coherent attacker.*

Contributions. Each is stated with its measured result; all comparisons use 20 paired seeds and pre-registered decision rules.

1) *Measured noise dominance.* At $\varepsilon = 1.86$ an honest client update has norm 41.4, against 1.06 for the server's clean update, and cosine 0.0075 with it. The cosine falls monotonically from 0.086 without DP to 0.0033 at $\varepsilon = 0.79$ (Fig. 1).
2) *Allocation: placement matters, learning does not.* Round sensitivity rises in late rounds in 7 of 9 tested configurations and is flat in none. A one-parameter decaying schedule beats uniform by $+0.0139$ mAP@0.5 ($+23.5\%$, 16/20 seeds) at $\varepsilon = 1.86$. A DDPG agent collapses (0.000), and a TD3+GRU agent ties uniform (-0.0005). Bayesian optimisation (BO) and a TD3 agent constrained to $\pm 20\%$ around the theory schedule tie the fixed schedules. Neither of two server-side feedback signals carries usable information (Section V-F).
3) *More rounds do not help at fixed ε.* At $\varepsilon = 1.86$ the tuned schedule scores 0.073 / 0.086 / 0.057 / 0.040 at 12 / 25 / 50 / 100 rounds. Over the same range the no-DP model rises to 0.33–0.34, so the private model's share of it falls from 29% to 12%.
4) *Trust-based robustness fails under DP and is only partly repairable.* FLTrust loses to FedAvg at every ε from 30.16 to 0.79, including without attack (-75% at $\varepsilon = 1.86$). A noise-aware norm target (Variant N) recovers FedAvg-level accuracy at $\varepsilon = 1.86$. An attacker who knows the rule drives it to 0.0005 mAP@0.5, a loss against FedAvg (-0.013, 1/18 wins). Capping any client's trust weight at 1/8 holds the attackers at exactly 25% of the weight, but the model still collapses (0.0058).
5) *An auditable negative-results record.* The record holds every pre-registered prediction that failed, one retraction (an FL$\times$DP interaction whose sign reversed once the clipping bound was calibrated), and every discarded implementation (Section VII).

Scope and limitations. All security results and the main

allocation results use one dataset (Anti-UAV infrared [8]). Allocation is replicated on VisDrone [9] and CIFAR-10 [10]. Privacy is frame-level, and consecutive frames of one recording are correlated, so the guarantee for a whole recording is weaker than the stated ε. The trainable model is small (65,221 parameters on a frozen backbone), and absolute accuracy on the held-out test sessions is low (0.059 mAP@0.5 for uniform at $\varepsilon = 1.86$). The adaptive attacker is omniscient. These limits are detailed in Section IX.

II. Related Work

DP training and accounting. DP-SGD [2] clips per-sample gradients and adds Gaussian noise. Rényi DP [11] and concentrated DP [12] give tight composition, and Opacus [13] implements Poisson-subsampled RDP accounting, which we use throughout. User-level DP for FL was introduced by McMahan et al. [14]; our unit is the frame, which is weaker (Section IX). Hyper-parameter tuning itself leaks privacy [15]; we do not account for it (Section IX).

Time-adaptive privacy spending. Kiani et al. [3] propose "spend-as-you-go": clients save budget early by sampling less, then spend it later. They report gains over uniform DP-FedAvg at client-level $\varepsilon = 10$–30 on MNIST, FMNIST and Adult, averaged over three runs. Our results *agree* with their direction (less noise late is better) and do not contradict their numbers, which come from a different regime. We add three things: a direct measurement of why late rounds matter (a noise-pulse test); paired, corrected statistics; and tests of learned and searched schedules, which they do not evaluate. Their rule, implemented in our setting, is the worst arm on CIFAR-10 but wins on VisDrone (Section V).

Learned allocation. Reinforcement-learning agents that set per-round noise use actor-critic methods such as DDPG [16] and TD3 [17]. We test both, with the agents trained only on public data so that training spends no client privacy.

Byzantine-robust FL and attacks. Krum [4] and the coordinate-wise median [5] bound the influence of outliers. FLTrust [6] bootstraps trust from a small clean root dataset at the server: each update is weighted by the ReLU of its cosine with the server's root update and rescaled to the root update's norm. We use three standard attacks: ALIE [18], inner-product manipulation (IPM) [19], and Fang et al.'s full-knowledge attack [20].

DP and robustness together. Guerraoui et al. [7] show that naively composing DP noise with Byzantine-resilient aggregation degrades with the model dimension. Our measurements are an empirical counterpart in a federated detector. The trust signal FLTrust relies on shrinks with noise exactly as the dimension argument predicts (Section III).

Where existing work falls short, and what we add. No prior study we know of measures, in one federated task at $\varepsilon \leq 3$, with paired and corrected statistics: (i) whether allocation gains need learning; (ii) whether more rounds close the DP gap; and (iii) whether a trust-based defence survives both the DP noise and an attacker who knows the defence. We do not claim that FLTrust is flawed in its own (noise-free) regime. There it tied FedAvg and beat it under the Fang attack, as its authors report.

TABLE I
Notation.

N, R, K, B	clients (8), rounds, local steps (20), batch (64)
σ_t	noise multiplier in round t (the schedule)
C	clipping bound; C^* is its calibrated value
D	trainable parameters (65,221)
v_t	per-coordinate noise variance of an update, Eq. (1)
a_t	sensitivity of final accuracy to noise in round t
ρ_t	privacy spent in round t (zCDP units, Prop. 1)
g_0	server root update (FLTrust), clean
κ	cap on any client's normalised trust weight (Sec. VI-C)

III. Preliminaries and Theory

A. System model

There are $N = 8$ clients and R rounds. In round t every client starts from the global model θ_t and runs $K = 20$ local steps of DP-SGD: Poisson sampling at rate $q_i = B/n_i$ with $B = 64$, per-sample clipping at bound C, and Gaussian noise of standard deviation $\sigma_t C$. The optimiser is SGD with learning rate $\eta = 0.03$ and momentum $\mu = 0.9$. Client i sends

$$\Delta_{i,t} = s_{i,t} + \xi_{i,t}, \qquad \xi_{i,t} \sim \mathcal{N}(0, v_t I_D), \tag{1}$$
$$v_t = \Big(\tfrac{\eta \sigma_t C}{B}\Big)^2 \sum_{j=1}^{K} \Big(\sum_{k=0}^{K-j} \mu^k \Big)^2,$$

where $s_{i,t}$ is the clipped-gradient (signal) part, $D = 65{,}221$ is the number of trainable parameters, and v_t is the per-coordinate noise variance accumulated through momentum. FedAvg sets $\theta_{t+1} = \theta_t + \sum_i (n_i/n)\Delta_{i,t}$. Privacy is accounted per client with the Opacus RDP accountant and reported as the *worst client's* ε at $\delta = 10^{-5}$. The protected unit is one frame. Table I lists the symbols.

B. Allocation: where the noise should go

Assumption 1 (Separable sensitivity)**.** *The excess loss of the final model is, to first order, $\mathcal{L} \approx \mathcal{L}_0 + \sum_t a_t \sigma_t^2$ with $a_t \geq 0$ not depending on the schedule.*

Assumption 2 (Additive cost)**.** *Round t spends privacy $\rho_t = c/\sigma_t^2$ for a constant c, and the budget is $\sum_t \rho_t = \rho$.*

Assumption 2 is exact for the non-subsampled Gaussian mechanism under zCDP [12], and only approximate under Poisson subsampling.

Proposition 1. *Under Assumptions 1–2, the schedule minimising $\sum_t a_t \sigma_t^2$ subject to the budget is $\rho_t = \rho\sqrt{a_t}/\sum_s \sqrt{a_s}$. Uniform allocation is optimal if and only if a_t is constant.*

Proof. $\sum_t a_t \sigma_t^2 = c \sum_t a_t/\rho_t$. By Cauchy–Schwarz, $\big(\sum_t \sqrt{a_t}\big)^2 = \big(\sum_t \sqrt{a_t/\rho_t}\sqrt{\rho_t}\big)^2 \leq \big(\sum_t a_t/\rho_t\big)\big(\sum_t \rho_t\big)$, so $\sum_t a_t/\rho_t \geq (\sum_t \sqrt{a_t})^2/\rho$. Equality holds iff $\sqrt{a_t/\rho_t} \propto \sqrt{\rho_t}$, i.e. $\rho_t \propto \sqrt{a_t}$. With constant a_t this is uniform. □

The theory schedule A3 (Section IV) uses only the *shape* $\rho_t \propto \sqrt{a_t}$. It then rescales the whole schedule with the exact accountant, so the budget is always met exactly even where Assumption 2 is inexact.

When the result does not apply. Assumption 1 is the one most likely to fail. (i) We measure a_t by perturbing

one round of a *uniform* schedule. If the schedule changes the training trajectory, a_t changes with it. (ii) a_t is estimated with noise. Negative estimates must be floored, and a floored round receives the maximum noise. This is exactly where A3 fails (Section V). (iii) Clipping fixes each round's signal contribution at most C, so the gradient norm itself cannot drive a_t (it is clipped in every round; Section V). Any variation in a_t must come from how late-round noise propagates to the final model.

C. Robust aggregation: the honest signal under noise

Proposition 2 (Noise dominance)**.** *For a fixed unit direction $\hat{g}$ and an update as in Eq.* (1) *with D large, $\|\Delta_{i,t}\|^2 \approx \|s_{i,t}\|^2 + Dv_t$ and $\cos(\Delta_{i,t}, \hat{g}) \approx \langle s_{i,t}, \hat{g}\rangle / \sqrt{\|s_{i,t}\|^2 + Dv_t}$.*

Proof sketch. $\|\xi\|^2/D \to v_t$ and $\langle \xi, \hat{g}\rangle \sim \mathcal{N}(0, v_t)$, which is $O(\sqrt{v_t})$ against $\|\xi\| = O(\sqrt{Dv_t})$. The cross term $\langle s, \xi\rangle$ is $O(\|s\|\sqrt{v_t})$, negligible against Dv_t. □

Two consequences follow. First, FLTrust weights by this cosine and rescales every update to $\|g_0\|$. When $Dv_t \gg \|s\|^2$, the useful component left in each rescaled update is about $\|g_0\| \cos(\Delta, g_0)$, which tends to 0: the global step shrinks. Rescaling instead to the norm an honest update is known to have, $N_t = \sqrt{\|g_0\|^2 + Dv_t}$ ("Variant N"), preserves the step. Second, and only as a post-hoc reading of Section VI-C: an attacker who sends an update of norm N_t that is pure signal, with aggregation weight w, contributes a coherent push of size about wN_t. The honest clients jointly contribute about $(1-w)\|\bar{s}\|$, because their noise averages away. The attacker dominates once $w \gtrsim \|\bar{s}\|/(N_t + \|\bar{s}\|)$. With the measured norms at $\varepsilon = 1.86$ (honest 41.4; root update 1.06 as a proxy for $\|\bar{s}\|$), that threshold is about 2.5%. This is far below any per-client weight cap $\geq 1/N = 12.5\%$.

D. Predictions stated before the results

Every phase was pre-registered before its first run, with decision rules and predictions committed to version control. The predictions most relevant to the central claim were: **(H1)** if a_t is not flat, a fixed schedule with less noise in late rounds beats uniform at equal ε; **(H2)** a feedback-driven agent can beat the best fixed schedule only if the server-side signal predicts the outcome beyond the schedule; **(H3)** at fixed ε, more rounds plateau (summed signal-to-noise $\sqrt{R}/\sigma$ = 1.73, 1.85, 1.91, 1.94 for R = 12, 25, 50, 100); **(H4)** under DP the honest trust score is small (< 0.2), so FLTrust's weighting is driven by noise; **(H5)** a weight cap prevents the adaptive attacker's collapse. H5 failed (Section VII).

IV. Experimental Design

Data and leakage-free split. The Anti-UAV infrared training release [8] has 200 folders, 231,557 frames and 216,210 frames with a target. Folders are segments of longer recordings. We map every folder to its recording session (89 sessions) and split *by session*, never by frame or folder. The dataset's official train/validation split is not used: 23 of its 89 sessions appear in both, a leak that a file-level duplicate check misses. Sessions are assigned, stratified by background environment (sky, cloud, building, terrain), to five disjoint pools: clients 58 sessions, public proxy 11, test 12, validation 6, root 2. The proxy pool serves as public data for agent training, for calibrating the clipping bound, and for server-side signals. The root pool is FLTrust's clean server data (1,300 frames after stride-5 sampling; it covers only sky and terrain). The 8 clients are 2 per environment, giving non-IID data by background (client sizes 1,300–6,162 frames). Every pool uses every fifth frame.

TABLE II
Configuration shared by all main experiments.

Clients / rounds	8 / $R = 12$ (also 25–100, Sec. V-G)
Local training	20 DP-SGD steps/round, $B = 64$, Poisson
Optimiser	SGD, lr 0.03, momentum 0.9
Clipping	per-seed C^* from public proxy data
Accountant	Opacus RDP, $\delta = 10^{-5}$, worst-client ε (frame-level unit)
Trainable parameters	65,221 (frozen MobileNetV3-Small)
Aggregation	FedAvg; in Sec. VI also FLTrust, median, Krum, variants L, N
Evaluation	test pool (12 sessions), mAP@0.5
Seeds	20 paired (200–219), common random numbers

TABLE III
Operating points. All schedules at an operating point are rescaled to the same worst-client ε (to within 1%).

Point	Rounds R	Uniform σ	Worst-client ε ($\delta = 10^{-5}$)
P (primary)	12	2	1.86
S1 (longer)	25	2	2.72
S2 (tighter)	12	4	0.79

Model. We use a MobileNetV3-Small backbone [21] (ImageNet weights), frozen through block 8, and a 65,221-parameter centre-point detection head. The metric is single-class mAP@0.5 computed with pycocotools [22]. A trivial predictor scores 0.000.

DP configuration. The clipping bound C^* is calibrated per seed at zero privacy cost: it is the median per-sample gradient norm after two rounds' worth of noise-free steps on the public proxy pool (mean 5.15–5.50). Table II gives the full configuration and Table III the operating points.

Allocation arms. All are rescaled to the same worst-client ε; σ_t is clipped to $[0.3, 20]$.

- A1: uniform.
- A2: exponential decay, $\sigma_t \propto e^{-k(t-1)/(R-1)}$, with k tuned on the validation pool over 5 seeds ($k = 0.5$ at P).
- A3: theory, $\rho_t \propto \sqrt{a_t}$ (Prop. 1), with a_t taken from the noise-pulse test.
- A4: DDPG [16] that sees only the round index and the remaining budget.
- A5: TD3 [17] with a GRU over rounds. Each round it adds three server-side signals computed on public proxy data (proxy loss, proxy gradient norm, change in proxy loss).
- A6: Kiani et al.'s spend-as-you-go rule [3], realised through σ rather than client subsampling.

- A7: Bayesian-optimisation search over full schedules, with compute matched to A5.
- A5-relative: A5's learner, restricted to adjusting A3's σ_t by $\pm 20\%$ while the remaining rounds are re-planned so the budget is met exactly.

Agents train for 300 episodes in a simulator built from the public proxy pool only. The simulator has 8 pseudo-clients of exactly the real clients' sizes, so its σ equals the real one. Agent training therefore spends no client privacy. The deployed checkpoint is the one with the best simulator evaluation reward, as fixed in advance.

Attacks and aggregators. Two of 8 clients (c3, c6) are malicious; they hold 17.1% of the training data and so of FedAvg's weight.

- ALIE [18] sends $\mu - zs$, with μ and s the mean and standard deviation of the honest updates and $z = 0.319$.
- IPM [19] sends -0.5μ.
- Fang [20] uses the full-knowledge trimmed-mean/median variant.
- The *trust-capture* attack knows the defence's rule and the root update g_0. It sends an update of norm N_t that has cosine c with g_0 and is otherwise opposed to the honest mean, with c chosen to maximise the harmful step.

The aggregators are FedAvg, FLTrust [6] (root update clipped at C^*, no noise), the coordinate-wise median [5], Krum [4] ($f = 2$), and our variants L (noise-corrected cosine) and N (noise-aware norm target).

Statistics. For every comparison we report the paired mean difference over 20 seeds, its 95% t-interval, wins and losses, and a p-value. Comparisons are grouped into pre-registered families with Holm correction [23]. A result is a WIN or LOSS if Holm rejects at $\alpha = 0.05$, and otherwise a TIE. A tie is EQUIVALENT if the 95% CI lies within ± 0.02 mAP@0.5. Seed s gives every arm the same initialisation, C^*, data order and noise stream (common random numbers). Per-seed differences for the headline comparisons are in Fig. 6.

Pre-registration, logging and reproduction. Each phase's design, decision rule and predictions were committed before its first run. Changes were made only by dated amendments written before the affected runs. Every number traces to a result file that records the code commit, machine, seed and configuration. Before each new grid, the new code reproduced stored earlier runs bit-for-bit, for example 80/80 FedAvg runs of the allocation study when re-run inside the security code.

Design choices that were forced. Each was forced by an earlier result rather than chosen.

- The batch and clipping bound: at $B = 256$ with the uncalibrated $C = 45.8$ the model collapsed at the weakest noise level.
- The calibrated C^*: the fixed $C = 45.8$ clipped only 0.3–0.5% of samples after round 1. Adaptive per-round clipping lost to it at 6 noise levels (e.g. -0.112 [-0.169, -0.055], 2/10).
- The simulator: the pre-registered one, with one recording per pseudo-client, could not train even without noise (reward 0.002–0.004).
- FLTrust's clipped root update: the unclipped root update first specified diverged.

Hardware. All runs used one NVIDIA GeForce RTX 5090 (PyTorch 2.7.1, CUDA 12.8), with up to three runs in parallel and no driver power cap. Software throttling was used instead; it inserts sleeps only, so results are numerically unaffected. A 12-round run took about 20–30 s and a 100-round run about 3 min. Agent training took 1–8 hours of wall-clock time per agent on the shared GPU.

V. Results I: Noise Dominance and Allocation

A. Sanity checks

The setup behaves as expected.

- A single non-private client reaches 0.584 ± 0.005 validation mAP@0.5; the trivial predictor scores 0.000.
- Accuracy falls monotonically with σ (Spearman -1.000).
- Clipping alone costs nothing ($\sigma = 0$: 0.576 vs 0.573 non-private).
- With the calibrated C^*, a central model falls from 0.562 ($\sigma = 0$) to 0.066 ($\sigma = 2$) on validation, while federated training with DP at $\sigma = 2$ keeps 0.286.

The gradient norm falls monotonically within a run (e.g. 26.8 $\rightarrow$ 9.4 at $\sigma = 2$). It stays above $C^* \approx 5.5$ in every round, so nearly every per-sample gradient is clipped. This is the setting assumed in Section III.

B. Honest DP updates are mostly noise

At $\varepsilon = 1.86$ an honest client update has norm 41.4. The server's clean root update has norm 1.06, and the cosine between them averages 0.0075. Without DP the two norms are 0.74 and 0.75 and the cosine is 0.086. Across privacy levels the cosine falls monotonically: 0.086, 0.027, 0.015, 0.0075, 0.0033 at σ = 0, 0.5, 1, 2, 4 (Fig. 1a). This is Proposition 2: with $D = 65{,}221$ the noise term Dv_t dominates. What to notice: at the privacy levels that matter, a client's update points almost at random.

C. Late-round noise hurts more

Round sensitivity a_t is not flat in any tested configuration and always rises late. The noise-pulse test doubles σ in a single round t and measures the drop in final validation mAP@0.5 against a paired baseline, with 20 seeds per round. It was run on 3 variants (12 or 25 rounds; 65k or 116k trainable parameters) $\times$ 3 noise levels, 3,120 runs in total. 7 of 9 cells vary significantly (slope and early-vs-late CIs exclude zero). The other 2 are not identifiable, because the larger model sits near the floor. None is flat. At P the drop rises from $+0.014$ (round 1) to $+0.187$ (round 12) on a baseline of 0.258. The slope is $+0.0100$ per round [$+0.0088$, $+0.0112$]. The same shape appears on CIFAR-10 and VisDrone (Fig. 2). At 25 rounds the sensitivity is flat for rounds 1–20 and concentrated in the last $\sim$5. What to notice: the last round alone accounts for most of the sensitivity on all three datasets.

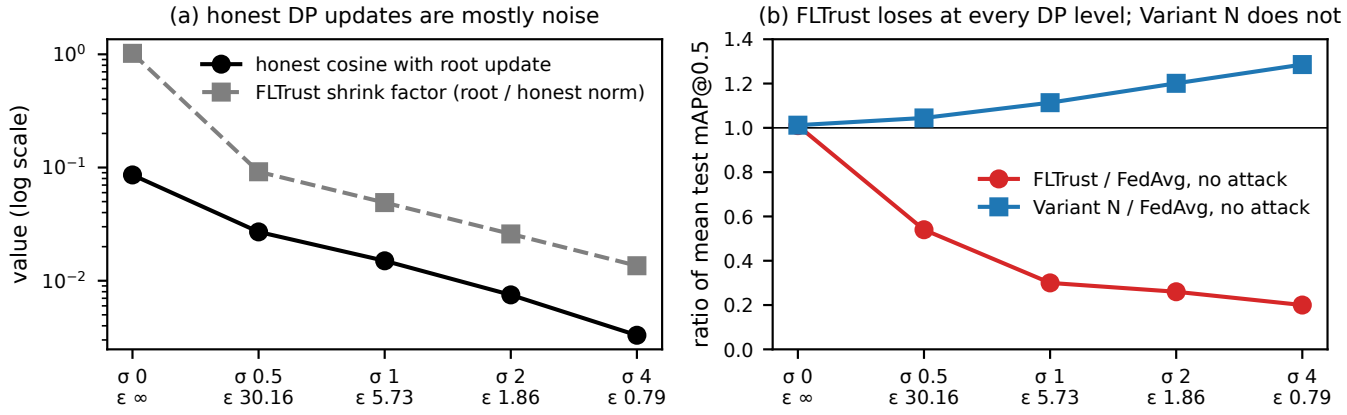


Fig. 1. Noise dominance across privacy levels (no attack, 20 seeds, uniform schedule). (a) Mean cosine between an honest client update and the server's clean root update, and FLTrust's shrink factor (root-update norm / honest-update norm), log scale. (b) Ratio of mean test mAP@0.5 to FedAvg for FLTrust and for Variant N. What to notice: the cosine and the shrink factor both fall by over an order of magnitude as ε tightens, and FLTrust's accuracy follows them (Spearman $+1$ over the five levels). Variant N stays at or above FedAvg in mean.

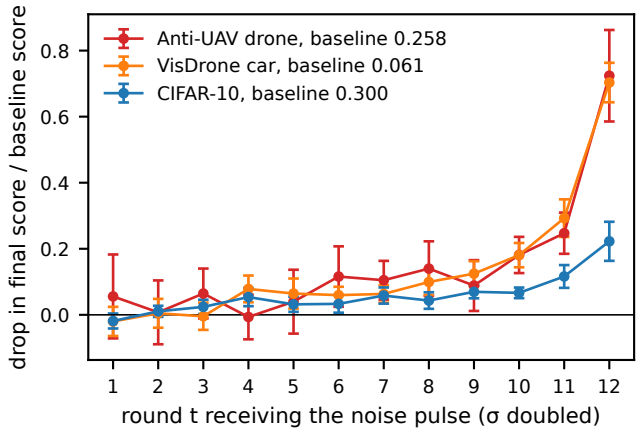


Fig. 2. Round sensitivity a_t: drop in final validation score when σ is doubled in one round, divided by the baseline score; 20 paired seeds per round, 95% CI; $\varepsilon = 1.86$, 12 rounds. What to notice: all three datasets rise late; the drone and VisDrone profiles almost coincide, while CIFAR-10's rise is about three times weaker relative to its baseline.

D. Main comparison

At P, the tuned decaying schedule (A2) beats uniform by $+0.0139$ mAP@0.5 [$+0.0076$, $+0.0202$] on 16/20 seeds, a relative gain of 23.5%. The theory schedule (A3) gains $+0.0144$ [$+0.0026$, $+0.0262$], also on 16/20 seeds. Neither learned agent helps (Table IV). The feedback agent A5 ties uniform (-0.0005 [-0.0124, $+0.0115$], equivalent within ± 0.02). It loses to A2 (-0.0144, 5/15, Holm $p = 0.011$) and to A3 (-0.0149, 4/16, Holm $p = 0.0013$). The blind agent A4 collapses to 0.000 on all 40 test runs at P and S2: it front-loads spending and is forced to $\sigma = 20$ in late rounds (Fig. 3). Under the pre-registered primary family the verdict is that learned allocation does not help. The A2 and A3 comparisons with uniform were registered as secondary, so they are unadjusted.

At the secondary points the picture is weaker. At S2 ($\varepsilon = 0.79$), A2 and A3 again beat uniform ($+0.0111$ and $+0.0071$, 16/4 each), near the floor. At S1 (25 rounds) the decay gain is not significant ($+0.0055$ [-0.0005, $+0.0114$]). There the theory schedule loses to uniform (-0.0248 [-0.0371, -0.0125], 3/17). Its per-round a_t estimates are noisy, many are floored, and the floored rounds receive the capped $\sigma = 4.25$.

E. Generalisation beyond the drone data

The direction replicates on VisDrone, but the size of the gain does not transfer reliably.

- *VisDrone (car, tile-level).* Same client sizes, so $\varepsilon = 1.86$. A2 beats uniform by $+0.0037$ [$+0.0020$, $+0.0054$], 17/3, Holm $p = 0.0006$, a $+10.5\%$ relative gain. The pre-registered "drone-like" line was 10%, so this classification is narrow (CI of the relative gain $+5.6\%$ to $+15\%$). A3 ties uniform, again because its early-round a_t estimates are floored. Kiani et al.'s rule (A6) wins ($+0.0032$, 17/3).
- *CIFAR-10.* The same a_t shape appears, but the gain is $+0.0050$ accuracy (16/4, Holm $p = 0.088$), $+1.7\%$ relative. All primary comparisons are equivalent within ± 0.02, and A6 is the worst arm (loses to A2 on 19/20 seeds).
- *Pascal VOC [24].* Stopped at the pre-registered quality gate: the frozen-trunk head reaches only 0.058–0.074 mAP@0.5 without noise under three configurations (bar 0.104). An oracle check (0.990) ruled out a scoring bug.

A non-flat a_t is therefore necessary for a benefit but does not set its size.

F. Nothing for learning to add

Neither server-side signal tested tells a policy anything the schedule does not.

- *Proxy loss.* There is large between-seed spread early (up to 28.6% of total training progress under A3), but it does not persist. Where the spread is large (rounds 1–5), a seed's loss barely predicts its final loss ($\rho \approx 0$). Where it persists (rounds 8–11, ρ 0.5–0.75), the spread is 4–8%.

TABLE IV
ALL ALLOCATION METHODS ON THE ANTI-UAV DRONE TASK. MEAN TEST MAP@0.5 OVER 20 PAIRED SEEDS. ALL METHODS AT AN OPERATING POINT SPEND THE SAME WORST-CLIENT ε (MATCHED TO ± 0.003). "GENUINE" MEANS SIGNIFICANTLY BETTER THAN UNIFORM *and* THAN THE BETTER OF A2/A3 (FIXED IN ADVANCE: A3 AT P, A2 AT S1/S2), HOLM-ADJUSTED WITHIN EACH PHASE'S FAMILY.

Method	P (ε 1.86)	S1 (ε 2.72)	S2 (ε 0.79)	Genuine improvement?
A1 uniform	0.0592	0.1050	0.0143	reference
A2 exponential decay (tuned)	0.0731	**0.1105**	**0.0254**	beats uniform at P, S2 (16/4); S1 n.s. (14/6)
A3 theory, $\rho_t \propto \sqrt{a_t}$	**0.0736**	0.0802	0.0213	beats uniform at P, S2 (16/4); *loses* at S1 (3/17)
A4 DDPG, blind	0.0000	0.1078	0.0000	no: collapses at P and S2 (0/20)
A5 TD3+GRU, feedback	0.0587	0.0981	0.0113	no: loses to A2/A3 at P (5/15, 4/16)
A7 Bayesian optimisation	0.0747	0.1060	0.0129	no: beats uniform at P, ties A2/A3; loses to A2 at S2 (1/19)
A5-relative ($\pm 20\%$ around A3)	0.0708	0.0921	0.0208	no: ties A2/A3 at P, S2; loses to A2 at S1 (3/17)

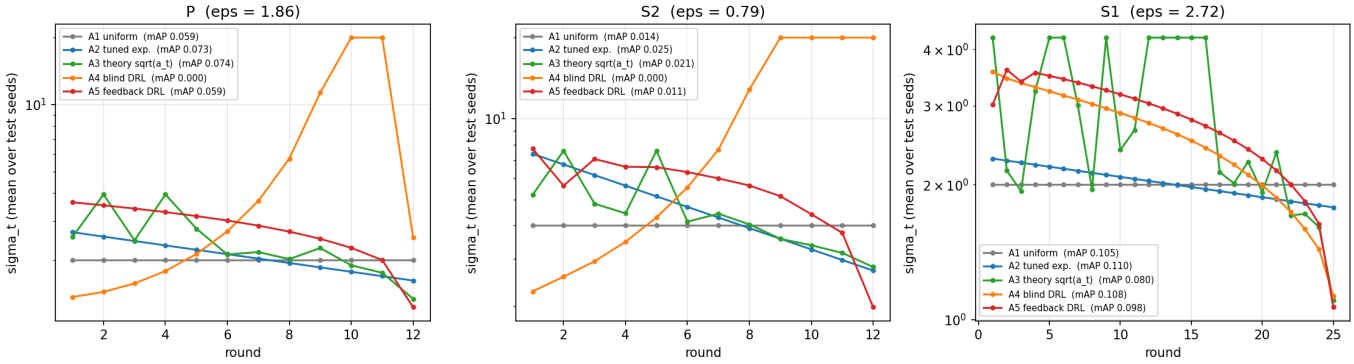


Fig. 3. Deployed noise schedules σ_t (mean over the 20 test seeds; log scale) at P, S2 and S1, with each arm's test mAP@0.5. What to notice: A2 and A3 decay gently; A4 front-loads spending and is forced to $\sigma = 20$ late (0.000 mAP@0.5); A5 sits at the most back-loaded schedule its action range allows (pinned at the minimum action) and does not beat uniform on test; at S1 A3 inherits noisy a_t estimates (spikes at the cap $\sigma = 4.25$). Regenerate as vector for the final version.

- *Gradient norm (3-round rolling mean).* It does not predict final validation mAP@0.5 (best $|\rho| = 0.30$ in rounds 1–8 under A3). It moves in lockstep with the round's own noise dose ($r = +0.68$ [0.51, 0.85]).

A5's deployed schedule was identical under every attack and aggregator in Section VI (max σ difference 2×10^{-6}): it ignores its inputs.

Two alternatives with the same compute also fail to beat the fixed schedules.

- *Bayesian optimisation (A7).* It beats uniform at P ($+0.0155$ [$+0.0050$, $+0.0261$], Holm $p = 0.019$) but ties A3 ($+0.0011$) and A2 ($+0.0016$). Its best simulator scores were mostly the maximum of noisy trials: on fresh seeds the S2 pick was already worse than uniform in the simulator (-0.050).
- *A5-relative.* It ties A3 (-0.0028, EQUIVALENT) and A2 at P, and loses to A2 at S1 (-0.0184, 3/17). Its training reward never rose at any operating point. Its adjustments are structured and do vary with feedback (between-seed sd 0.051 at P), but they do not improve accuracy.

G. More rounds at fixed privacy

At fixed $\varepsilon = 1.86$, more rounds do not close the gap to the no-DP model; past 25 rounds they widen it (Fig. 4). A2 scores 0.0731, 0.0861, 0.0573 and 0.0401 at 12, 25, 50 and 100 rounds; uniform follows the same curve. The no-DP model scores 0.250, 0.321, 0.340 and 0.330, so the private share of it falls from 0.29 to 0.12. In the pre-registered family:

- $25 - 12$: $+0.0131$, TIE.
- $50 - 25$: -0.0288 [-0.0457, -0.0119], 7/13, LOSS.
- $100 - 50$: -0.0172, 6/14, Holm $p = 0.056$, TIE.

The verdict is therefore "noise ceiling / plateau", not formally "declining". A2's advantage over uniform is constant: $+0.014$, $+0.012$, $+0.012$, $+0.013$, with 15–17 wins out of 20 at every R. Fixing ε forces the per-round σ up as R grows (uniform $2.00 \to 5.15$). The final proxy loss of the DP model rises with R (0.89, 1.11, 1.43), while the no-DP model reaches 0.43–0.45 at every R.

VI. RESULTS II: TRUST-BASED ROBUSTNESS UNDER DP

A. FLTrust fails at every privacy level

At $\varepsilon = 1.86$, FLTrust loses to FedAvg with no attack (-0.0441 [-0.0596, -0.0286], 2/18), a 75% loss of accuracy. It also loses under ALIE (0/20), IPM (1/19) and Fang (3/17). The classical rules also cost accuracy without attack: median -0.0214 (4/16) and Krum -0.0507 (2/18), both LOSS. Across privacy levels:

- FLTrust loses to FedAvg without attack at $\varepsilon = 30.16$ (-0.082, 0/20), 5.73 (-0.093, 0/20) and 0.79 (-0.011, 3/17).
- It loses under every attack except Fang at $\varepsilon = 0.79$, where it ties at the floor.
- Its accuracy relative to FedAvg tracks the honest cosine and the shrink factor exactly (Spearman $+1.00$ for each, over five levels; Fig. 1).

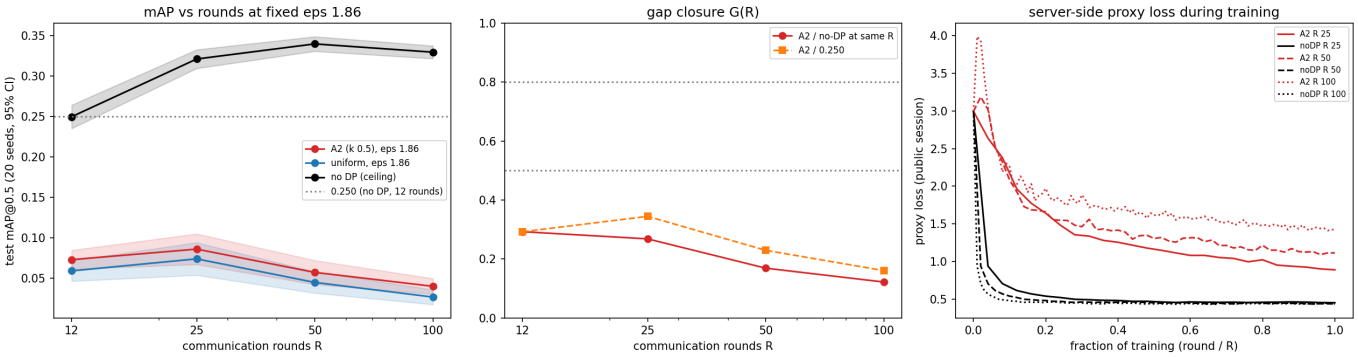


Fig. 4. Rounds at fixed worst-client $\varepsilon = 1.86$. Left: test mAP@0.5 (20 seeds, 95% CI) vs the number of rounds for A2, uniform, and no DP. Middle: gap closure $G(R)$ = A2 / no-DP at the same R. Right: server-side proxy loss during training. What to notice: the DP curves peak at 25 rounds while no DP keeps rising to 50, and the DP proxy loss ends higher the longer the run. Regenerate as vector for the final version.

- Without DP it ties FedAvg and beats it under Fang ($+0.045$, 16/4), as designed.

What to notice: the failure is specific to DP, and it grows as privacy tightens.

B. A noise-aware repair, and its limits

The two possible culprits can be separated. Correcting the cosine (Variant L) does not help; correcting the norm target (Variant N) does. Variant L still loses to FedAvg in all four conditions (-0.042, -0.036, -0.058, -0.023). Variant N rescales each update to $N_t = \sqrt{\|g_0\|^2 + Dv_t}$ instead of $\|g_0\|$ and keeps FLTrust's trust weights. At $\varepsilon = 1.86$ it ties FedAvg with no attack ($+0.012$ [-0.008, $+0.032$], 9/11), under ALIE ($+0.004$) and under IPM (-0.009), and beats it under Fang ($+0.026$ [$+0.008$, $+0.045$], 15/5). It beats plain FLTrust in all four conditions (17/3 to 20/0). The ties are not equivalences: the CIs reach $+0.032$ and -0.025. The no-attack mean gain comes largely from one seed (per-seed range -0.026 to $+0.131$; Fig. 6).

Variant N does not recover at every privacy level. Pre-registered family F8 gives:

- $\sigma = 0$ ($\varepsilon = \infty$): recovers; Variant N is identical to FLTrust there on 80/80 runs.
- $\sigma = 0.5$ ($\varepsilon = 30.16$): recovers.
- $\sigma = 1$ ($\varepsilon = 5.73$): loses under IPM (-0.030 [-0.052, -0.009], 6/14).
- $\sigma = 4$ ($\varepsilon = 0.79$): loses under ALIE (-0.011, 1/19, near the floor).

C. An adaptive adversary, and why a weight cap does not stop it

An attacker who knows Variant N's rule and the root update drives it from 0.071 to 0.0005 mAP@0.5. At three decimals 18 of 20 seeds score 0.000, and 16 are exactly zero. This is a LOSS against FedAvg under the same attack (-0.013 [-0.023, -0.003], 1/18), and below plain FLTrust (0.0055). Because Variant N gives every update the same norm, weight is decided by trust alone. Honest trust scores sum to only $\sim$0.04 per round, so an update with cosine $c \approx 0.3$–0.5 takes about 60–73% of the weight. The attack is strong in general: it also cuts FedAvg from 0.059 to 0.014.

Capping any client's normalised trust weight at $\kappa = m/8$ does not fix this. We tested $m = 2$, 1.5 and 1. The cap redistributes excess weight by water-filling and drops it when too few clients have positive trust. It works exactly as designed: the attackers' share is held at 2κ = 50%, 37.5% and 25%, down from 73%. Yet the model still collapses (Table V, Fig. 5).

- "Collapse prevented" (capped vs uncapped under the attack) ties at every margin.
- At $m = 1.5$ the capped system loses to FedAvg (0/19).
- A cap-aware attacker, who picks the smallest cosine that still reaches the cap, is no stronger.
- The cap costs nothing in the standard conditions (no LOSS), and at $m = 1$ it even improves robustness to ALIE ($+0.031$ [$+0.012$, $+0.050$], 15/5).

The post-hoc reading of Proposition 2 explains this. Once honest updates are mostly noise, a coherent update needs only $\sim$2.5% of the weight to dominate the aggregate's useful direction. FedAvg, which gives the attackers 17.1%, is itself badly hurt.

VII. Failed Predictions, Null Results and Retractions

Retraction: the FL×DP interaction. An early 2×2 experiment found that DP hurts far more inside FL than centrally (interaction -0.120 [-0.178, -0.062], 0/10). That run used a clipping bound $C = 45.8$ measured on the untrained model. After round 1 this bound clipped only 0.3–0.5% of samples, while the noise stayed sized for it. With the calibrated C^* the interaction reverses sign: $+0.135$, $+0.330$ and $+0.201$ at σ = 1, 2, 4, 10/10 seeds each. Under calibrated clipping, central DP-SGD suffers more than federated DP at equal σ; the federated clients have larger ε at the same σ, so the cells are matched on σ, not on ε. Every result in this paper uses C^*.

Failed pre-registered predictions (Table VI).

Null results reported as such:

- learned allocation (A4, A5) vs uniform;

TABLE V
PER-CLIENT TRUST-WEIGHT CAP ON VARIANT N AT $\varepsilon = 1.86$. MEAN TEST MAP@0.5, 20 PAIRED SEEDS; [SEEDS AT 0.000]. BOTTOM ROWS: HOLM-ADJUSTED TESTS WITHIN EACH MARGIN (PRE-REGISTERED FAMILY, 6 TESTS).

	Variant N, no cap	cap $2/8$	cap $1.5/8$	cap $1/8$	FedAvg
No attack	0.071	0.072	0.071	0.065	0.059
ALIE	0.044	0.051	0.066	**0.074**	0.039
IPM	0.070	0.068	0.079	0.081	0.078
Fang	0.070	0.067	0.073	0.074	0.044
Trust-capture	0.0005 [18]	0.0004 [19]	0.0000 [20]	0.0058 [14]	0.0137 [7]
Trust-capture, cap-aware	–	0.0003 [18]	0.0014 [12]	–	–
Attackers' weight share	73%	50%	37.5%	25%	17.1% (data)
Capped vs no cap, trust-capture	–	tie	tie	tie	–
vs FedAvg, trust-capture	-0.013, 1/18, LOSS	-0.013, tie	-0.014, 0/19, LOSS	-0.008, tie	–
Genuine fix?	–	**no**	**no**	**no**	–

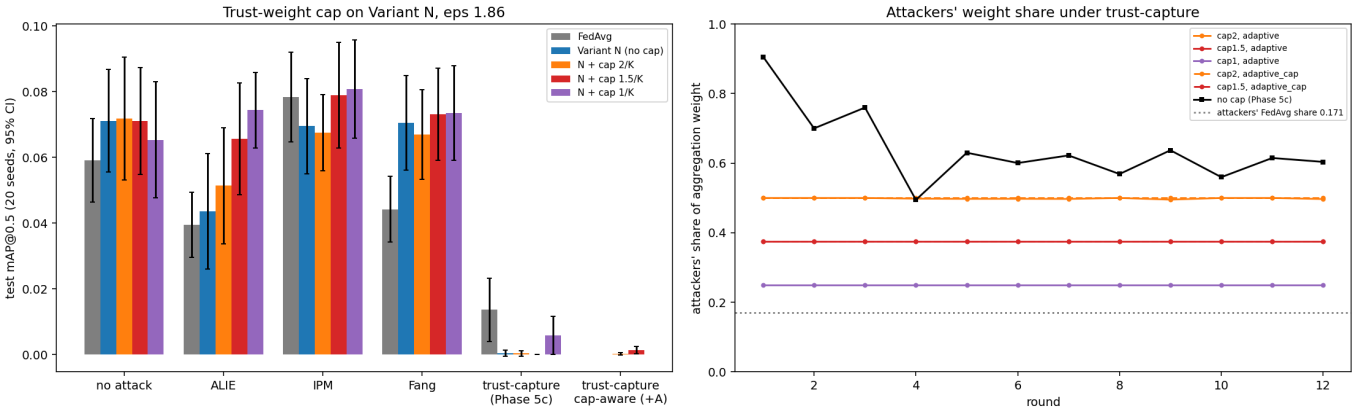


Fig. 5. Left: mean test mAP@0.5 (20 seeds, 95% CI) of FedAvg, Variant N and three trust caps under no attack, the three standard attacks, the trust-capture attack and its cap-aware variant. Right: the attackers' share of aggregation weight per round under trust-capture. What to notice: the caps hold the attackers' share flat at exactly 2κ, yet every capped bar under trust-capture is at the floor. Regenerate as vector for the final version.

TABLE VI
PRE-REGISTERED PREDICTIONS THAT FAILED. HELD PREDICTIONS ARE LISTED IN THE PER-PHASE REPORTS IN THE REPOSITORY.

Prediction	Outcome
Rounds: flat accuracy at fixed ε (H3)	held as "no gain", but accuracy *fell* past 25 rounds
A5-relative beats A5 at P	$+0.012$, Holm $p = 0.14$: tie
A5-relative ignores feedback (sd $<$ 0.02)	sd 0.051–0.085: it reacts
Variant N only partly recovers at $\varepsilon = 1.86$	failed in the good direction: full recovery
Variant N recovers at every ε	fails at $\sigma = 1$ (IPM) and $\sigma = 4$ (ALIE)
Trust-capture takes $\geq$ 85% of weight	$\sim$60–73%
Trust-capture hurts FedAvg much less	FedAvg 0.059 $\rightarrow$ 0.014
Weight cap prevents collapse (H5)	no margin does
Cap at $m = 2$ loses to FedAvg	tie (Holm $p = 0.064$)
Cap at $m = 1.5$ is a narrow fix	fails; LOSS vs FedAvg

- BO and A5-relative vs the fixed schedules;
- CIFAR-10 replication (equivalent);
- FedProx [25] (μ 0.01–0.3 tie at $\sigma = 1, 2$; $\mu = 1$ loses at $\sigma = 2$, -0.050, 1/10);
- the schedule's effect on attack damage under FLTrust (6/6 equivalent within ± 0.008).

Discarded implementations (all fixed by dated amendments before the reported runs):

- the agent action mapping first measured spend in a zCDP-style unit that understated the true cost under subsampling;
- the TD3+GRU critic target dropped the first observation;
- the first simulator could not train;
- an unclipped FLTrust root update diverged;
- a first A5-relative training run stalled on memory and was restarted from scratch with the same seeds.

Unexpected effect. IPM *helps* FedAvg under DP ($+0.019$, 16/4 seeds better). Its two updates are $-0.5\times$ the mean of six noisy updates, so they carry about $\sqrt{6}$ less noise. Replacing two noisy updates with them reduces the aggregate noise more than the reversed direction hurts. Without DP, IPM has no effect.

Closest prior method on its own benchmark. [MISSING: a reproduction of Kiani et al. [3] on MNIST/FMNIST/Adult at client-level DP. We implemented their rule only inside our setting (A6), via σ instead of client subsampling.]

VIII. DISCUSSION

One fact, two consequences. At $\varepsilon \leq 3$ with $D = 65{,}221$, each honest update is noise plus a faint signal (cosine 0.0075). For allocation, the accuracy cost of noise is set by where it lands: late noise reaches the final model with little time

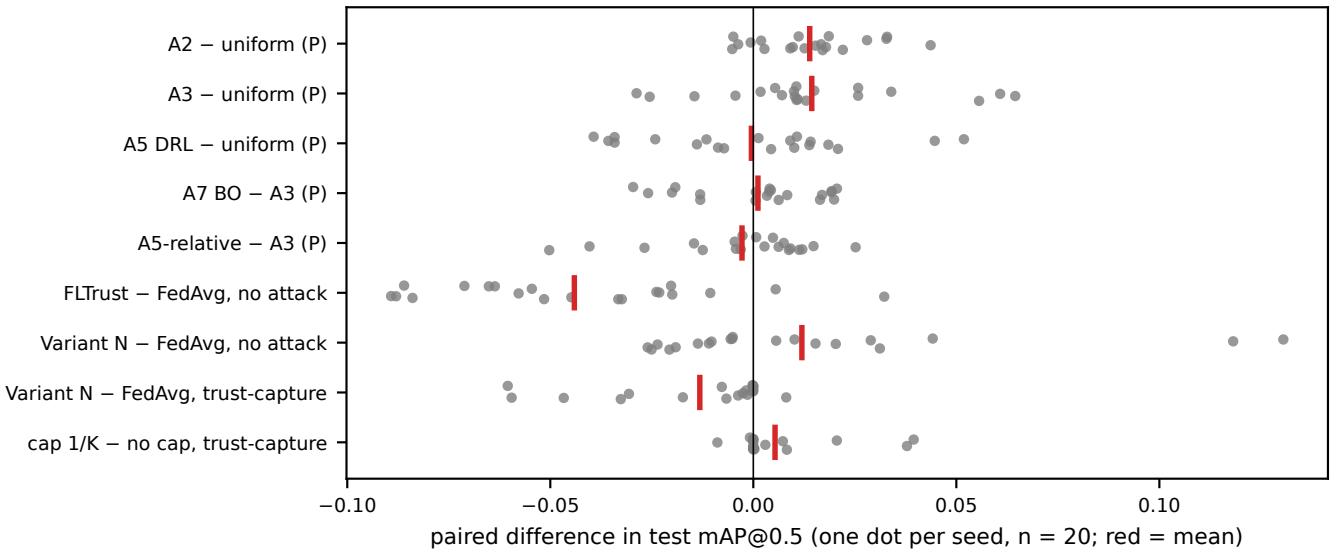


Fig. 6. Per-seed paired differences in test mAP@0.5 for the headline comparisons ($n = 20$ each, one dot per seed, red bar = mean; P unless stated). What to notice: the decay schedules' gain over uniform is consistent (16/20 seeds), the learned and searched schedules scatter around the fixed ones, and Variant N's mean gain without attack rests partly on one seed ($+0.131$).

to average out (Fig. 2). A fixed decaying schedule captures this. A feedback policy would need a signal that reveals more than the schedule already implies. The two server-side signals we tested do not: one is transient, the other tracks the noise just injected. For robustness, a defence that reads per-update geometry has almost nothing honest to read. FLTrust's norm equalisation then throws away 97% of every update. Once the norm is repaired, any coherent update dominates at a weight far below what per-client caps can enforce. The two halves meet at Proposition 2.

Practical recommendations.

1) Use a fixed, gently decaying noise schedule (k between 0.25 and 1, tuned on validation data) rather than uniform when privacy is strong and accuracy is off the floor: $+23.5\%$ on the drone task and $+10.5\%$ on VisDrone, at no privacy or training cost (Table IV).
2) Do not train a reinforcement-learning or search-based allocator for this regime. It cost 1–8 GPU-hours per agent and gave no gain, and the blind DDPG agent collapsed (Table IV).
3) Do not raise the round count to buy accuracy at a fixed ε. Beyond about 25 rounds it lowered accuracy (Fig. 4).
4) Do not deploy FLTrust unmodified under DP. Plain FedAvg was the best of FedAvg, FLTrust, median and Krum under every attack at $\varepsilon = 1.86$ (Section VI).
5) If a trust-based rule is used, rescale to the known DP norm (Variant N). Treat it as a utility fix only: it is broken by an attacker who knows the rule (Section VI-C).
6) Per-client weight caps are cheap and help against ALIE, but they are not a defence against a coherent adaptive attacker (Table V).

Relation to prior work. Our allocation direction agrees with Kiani et al. [3]. Their rule is the worst arm on CIFAR-10 in our setting and wins on VisDrone, so its value is dataset-dependent at strong privacy. Our robustness results are consistent with Guerraoui et al.'s dimension argument [7]. They do not contradict FLTrust [6] in its noise-free regime.

Tension. The repair that restores utility (equal norms) is what lets the adaptive attacker win. The original FLTrust's shrinkage was also a crude defence: plain FLTrust lost less to the trust-capture attack (0.0055) than Variant N did (0.0005). Under DP, utility and robustness pull the norm target in opposite directions.

IX. Limitations

- **Privacy unit.** Every ε is frame-level. Consecutive frames of a recording are highly correlated, so a recording-level guarantee is much weaker (group privacy).
- **Unaccounted tuning.** A2's k was tuned on the validation pool, and A3's a_t came from 3,120 training runs on client data. Their privacy cost [15] is not in the reported ε.
- **One primary dataset.** All security results and the main allocation results use Anti-UAV. Allocation replicates on VisDrone (narrowly) and only in direction on CIFAR-10. VOC could not be tested.
- **Low absolute accuracy and a validation-to-test shift.** Uniform scores 0.059 on test vs $\sim$0.29 on validation. The test sessions hold many tiny targets (median box 110 vs 621 px^2). Paired differences are unaffected, but absolute gains are small (+0.014 mAP@0.5).
- **Small trainable model** (65,221 parameters, frozen trunk). Effects may depend on D.
- **Test-pool reuse.** The same 12 test sessions serve all phases. Holm corrects within pre-registered families, not across phases.
- **Threat model.** Two fixed attackers (17.1% of the data). The root set covers 2 of 4 environments. The adaptive attacker is omniscient (it knows g_0); attackers who must

estimate g_0 are untested. No DP-native robust aggregators were compared.

- **Simulator.** Agents trained in a public-data simulator. Its pseudo-clients reuse near-duplicate consecutive frames.
- **Stand-ins.** Kiani et al.'s rule was realised through σ, not client subsampling. VisDrone mAP is tile-level for one class.

X. Open Problem

How can a server bound a coherent attacker's influence when honest updates are mostly noise? Weight-based defences cannot. By Proposition 2 the attacker needs only a few percent of the weight, below any per-client cap that still averages honest noise. Direction-based defences need the honest direction, which DP hides at the level of a single update. That is the purpose of DP. A defence would have to accumulate evidence across rounds (an attacker's push is consistent, honest noise is not), or obtain a server-side signal strong enough to stand in for the honest direction. Both conflict with what the privacy guarantee removes. We therefore expect the problem to be structural at strong privacy rather than a matter of tuning.

XI. Conclusion

We measured both privacy-budget allocation and trust-based robust aggregation on one differentially private federated drone detector at worst-client ε 0.79–2.72. The measurements used pre-registered rules, 20 paired seeds and Holm correction.

- A fixed decaying schedule gave $+0.0139$ mAP@0.5 ($+23.5\%$) over uniform at $\varepsilon = 1.86$. Learned, searched and constrained-learned schedules gave nothing more.
- More rounds at a fixed ε lowered accuracy past 25.
- FLTrust lost 75% of its accuracy to DP noise. Its noise-aware repair was driven to 0.0005 mAP@0.5 by a rule-aware attacker, and a 1/8 weight cap still left 0.0058.

At strong privacy an honest DP update is almost entirely noise; consequently, all the allocation gain we could measure comes from placing less noise in later rounds, which a fixed schedule already captures, while trust-based robust aggregation either drowns the honest signal or hands control to a coherent attacker. *Practical advice:* decay the noise with a fixed schedule, and keep FedAvg unless a defence has been tested against an attacker who knows it. *Scientific claim:* at strong privacy, what can be learned or defended in federated training is bounded by how little signal a single noisy update carries.

Data and Code Availability

Pre-registrations, code, per-run result files and analyses are in the `restart/` directory of the project repository (branch `restart`). [MISSING: public repository URL and dataset licence statement.]

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